

2006 级数学物理方程试题 标准答案

一. 填空

1. $u_{tt} - a^2 (u_{xx} + u_{yy}) = f(x, y, t)$

2. 三

3. $\frac{\partial u}{\partial t} - a^2 \left(\frac{1}{\rho} \frac{\partial u}{\partial \rho} + \frac{1}{\rho^2} \frac{\partial^2 u}{\partial \theta^2} + \frac{\partial^2 u}{\partial \rho^2} \right) = f$

4. $x^2 + x + a^2 t^2$

5. $u(m_0) = \frac{1}{4\pi a^2} \iint_{K_a} u \, ds$

6. $x^2 \frac{d^2 y}{dx^2} + x \frac{dy}{dx} + (x^2 - n^2) y = 0$

7. $x^n J_{n-1}(x)$

8. $y - 3x = C_1, \quad y + x = C_2$

9. $f(t) \rightarrow F(\omega) = \int_{-\infty}^{+\infty} e^{-i\omega t} f(t) \, dt$

10. $\iiint_{\Omega} (u \nabla^2 v - v \nabla^2 u) \, dV = \iint_{\Gamma} (u \frac{\partial v}{\partial n} - v \frac{\partial u}{\partial n}) \, dS$

二. 第二类齐次边界条件的波初问题.

令 $u(x, t) = X(x)T(t)$ 代入方程得.

$$X'' + \lambda X = 0$$

$$T'' + a^2 \lambda T = 0$$

代入边界条件得, $X(0) = X(l) = 0$

本征值问题:
$$\begin{cases} X'' + \lambda X = 0 \\ X(0) = X(l) = 0 \end{cases} \quad \dots \dots 3 \frac{1}{2}'$$

本征值: $\lambda_n = n^2$

本征函数: $X_n = \cos nx$. $n = 0, 1, 2, \dots \dots 5 \frac{1}{2}'$

$$T_n'' + a^2 n^2 T_n = 0 \Rightarrow T_n = C_n \cos nat + D_n \sin nat, \quad n = 1, 2, \dots$$

$$T_0'' = 0 \Rightarrow T_0 = C_0 + D_0 t.$$

$$\text{则 } U_n = [C_n \cos nat + D_n \sin nat] \cos nx \quad U_0 = C_0 + D_0 t.$$

$$\Rightarrow u = C_0 + D_0 t + \sum_{n=1}^{\infty} [C_n \cos nat + D_n \sin nat] \cos nx. \quad \dots \dots 8 \frac{1}{2}'$$

代入初始条件.

$$C_0 + \sum_{n=1}^{\infty} C_n \cos nx = 2x.$$

$$D_0 + na \sum_{n=1}^{\infty} D_n \cos nx = 0$$

$$\Rightarrow C_0 = \frac{1}{\pi} \int_0^{\pi} 2x dx = \pi.$$

$$C_n = \frac{2}{\pi} \int_0^{\pi} 2x \cos nx dx = \frac{4}{n^2 \pi} [(1-1)^n - 1]$$

$$D_n = 0$$

$\dots \dots 10 \frac{1}{2}'$

三. 特征方程: $(dy)^2 + 6dx dy + 8(dx)^2 = 0$ ----- 2分

特征线: $y + 4x = C_1$ $y + 2x = C_2$

特征变换: $\xi = y + 4x$ $\eta = y + 2x$

通解: $u = f_1(y + 4x) + f_2(y + 2x)$ ----- 5分

代入定解条件:
$$\begin{cases} f_1(4x) + f_2(2x) = 5-x \\ f_1'(4x) + f_2'(2x) = 0 \end{cases}$$

求导: $f_1(4x) + 2f_2(2x) = C$

$$\begin{cases} f_2(2x) = -5x + C \\ f_1(4x) = 25x - C \end{cases}$$

解得: $f_1(y) = 25 \frac{y}{4} - C'$

$f_2(y) = -5(\frac{y}{2}) + C'$ ----- 8分

$u = 25(x + \frac{y}{4}) - 5(x + \frac{y}{2})$ ----- 10分

四、解：对 y 进行 Laplace 变换，得

$$\begin{cases} \frac{d}{dx} [p U(x, p) - 2x^2] = \frac{2}{p^3} \\ U(1, p) = \frac{1}{p} \end{cases}$$

由 1#

$$U(x, p) = \int_0^{+\infty} u(x, y) e^{-py} dy \quad \dots\dots\dots 4 \text{ 分}$$

解之，得

$$U(x, p) = \frac{2}{p} x^2 + \frac{2}{p^4} x - \frac{1}{p} - \frac{2}{p^4} \quad \dots\dots\dots 8 \text{ 分}$$

对 U 作 Laplace 逆变换，得

$$u(x, y) = 2x^2 + \frac{1}{3} x y^3 - 1 - \frac{1}{3} y^3 \quad \dots\dots\dots 10 \text{ 分}$$

五. 解: 设 $M_0(x_0, y_0, z_0)$ 为半空间 $y > 2$ 内的任一点. 则 $M_1(x_0, 4-y_0, z_0)$ 为 M_0 关于平面 $y=2$ 的对称点. 故可设 Green 函数

$$G(M, M_0) = \frac{1}{4\pi} \left(\frac{1}{r_{MM_0}} - \frac{1}{r_{MM_1}} \right)$$

其中 r_{MM_0} 为 M_0 到 $M(x, y, z)$ 的距离,

r_{MM_1} 为 M_1 到 M 的距离. ----- 5 分

$$\begin{aligned} \frac{\partial G}{\partial n} \Big|_{y=2} &= - \frac{\partial G}{\partial y} \Big|_{y=2} \\ &= - \frac{1}{4\pi} \frac{\partial}{\partial y} \left\{ \frac{1}{[(x-x_0)^2 + (y-y_0)^2 + (z-z_0)^2]^{3/2}} \right. \\ &\quad \left. - \frac{1}{[(x-x_0)^2 + (y-(4-y_0))^2 + (z-z_0)^2]^{3/2}} \right\} \Big|_{y=2} \\ &= \frac{1}{2\pi} \frac{2-y_0}{[(x-x_0)^2 + (2-y_0)^2 + (z-z_0)^2]^{3/2}} \end{aligned}$$

----- 9 分

故所求方程... 解为

$$\begin{aligned} u(x_0, y_0, z_0) &= - \frac{1}{2\pi} \int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} \frac{2-y_0}{[(x-x_0)^2 + (2-y_0)^2 + (z-z_0)^2]^{3/2}} \\ &\quad \times \rho(x, z) dx dz \end{aligned}$$

----- 12 分

六. (12分) 求满足下面定解问题的解.

$$\begin{cases} u_t = u_{xx}, & 0 < x < \pi, \quad t > 0 \\ u(0, t) = 0, & u(\pi, t) = 5, \\ u(x, 0) = x^2. \end{cases}$$

解: 令 $u(x, t) = v(x, t) + w(x)$, 代入到方程有 (2分)

$$v_t = u_t = v_{xx} + w''(x).$$

边界条件

$$v(0, t) = u(0, t) - w(0) = -w(0),$$

$$v(\pi, t) = u(\pi, t) - w(\pi) = 5 - w(\pi),$$

要使得 $v(x, t)$ 满足齐次方程, 齐次边界条件, $w(x)$ 需满足

$$\begin{cases} w''(x) = 0, \\ w(0) = 0, \quad w(\pi) = 5, \end{cases}$$

$$\text{解得 } w(x) = \frac{5}{\pi} x, \quad \dots \dots \dots (4分)$$

而 $v(x, t)$ 满足

$$\begin{cases} v_t = v_{xx}, \\ v(0, t) = v(\pi, t) = 0, \\ v(x, 0) = x^2 - \frac{5}{\pi} x, \end{cases}$$

用分离变量法, $v(x, t) = X(x)T(t)$, 代入得 (6分)

$$X(x) T'(t) = X''(x) T(t),$$

$$\text{从而 } \frac{X''(x)}{X(x)} = \frac{T'(t)}{T(t)} = -\lambda,$$

$$\text{得 } T'(t) + \lambda T(t) = 0,$$

$$\begin{cases} X''(x) + \lambda X(x) = 0 \\ X(0) = X(\pi) = 0 \end{cases}$$

$$\text{解得特征值 } \lambda_n = n^2, \quad n=1, 2, \dots$$

$$\text{对应的特征函数 } X_n(x) = \sin nx, \quad n=1, 2, \dots$$

$$\text{由 } T_n'(t) + n^2 T(t) = 0, \quad \text{解得}$$

$$T_n(t) = C_n e^{-n^2 t}, \quad n=1, 2, \dots \dots \dots (9分)$$

$$\therefore u(x, t) = \sum_{n=1}^{\infty} C_n e^{-n^2 t} \sin nx,$$

将 $t=0$ 代入得

$$x^2 - \frac{5}{\pi}x = \sum_{n=1}^{\infty} C_n \sin nx, \text{ 解得 } \dots\dots (10 \text{ 分})$$

$$C_n = \frac{1}{n\pi} \left[\frac{4}{n^2} - 2\pi^2 + 10 \right] \cos n\pi - \frac{4}{n^3\pi}$$

$$(\text{或}) C_n = \frac{1}{n\pi} \left[\frac{4}{n^2} - 2\pi^2 + 10 \right] (-1)^n - \frac{4}{n^3\pi} \dots\dots (12 \text{ 分})$$

t. (6分) 证明下式

$$j_3(x) + 3j_0'(x) + 4j_0'''(x) = 0,$$

其中 $j_3(x)$, $j_0(x)$ 分别是 3 阶和 0 阶贝塞尔函数.

证明: 利用递推关系式

$$j_{n-1}(x) - j_{n+1}(x) = 2j_n'(x), \dots\dots (2 \text{ 分})$$

取 $n=1, 2$, 分别得

$$j_0(x) - j_2(x) = 2j_1'(x), \quad \textcircled{1}$$

$$j_1(x) - j_3(x) = 2j_2'(x), \quad \textcircled{2}$$

①式求导乘以 2 得

$$2j_0'(x) - 2j_2'(x) = 4j_1''(x), \dots\dots (4 \text{ 分})$$

与②相减得

$$j_1(x) - j_3(x) - 2j_0'(x) = 4j_1''(x).$$

利用 $j_0'(x) = -j_1(x)$, 得

$$j_3(x) + 3j_0'(x) + 4j_0'''(x) = 0. \dots\dots (6 \text{ 分})$$